

LAB 1 — ACTORS, USE CASES & UML USE-CASE DIAGRAM

Problem Statement: Blood Bank Inventory & Emergency Donor Matcher

Actors

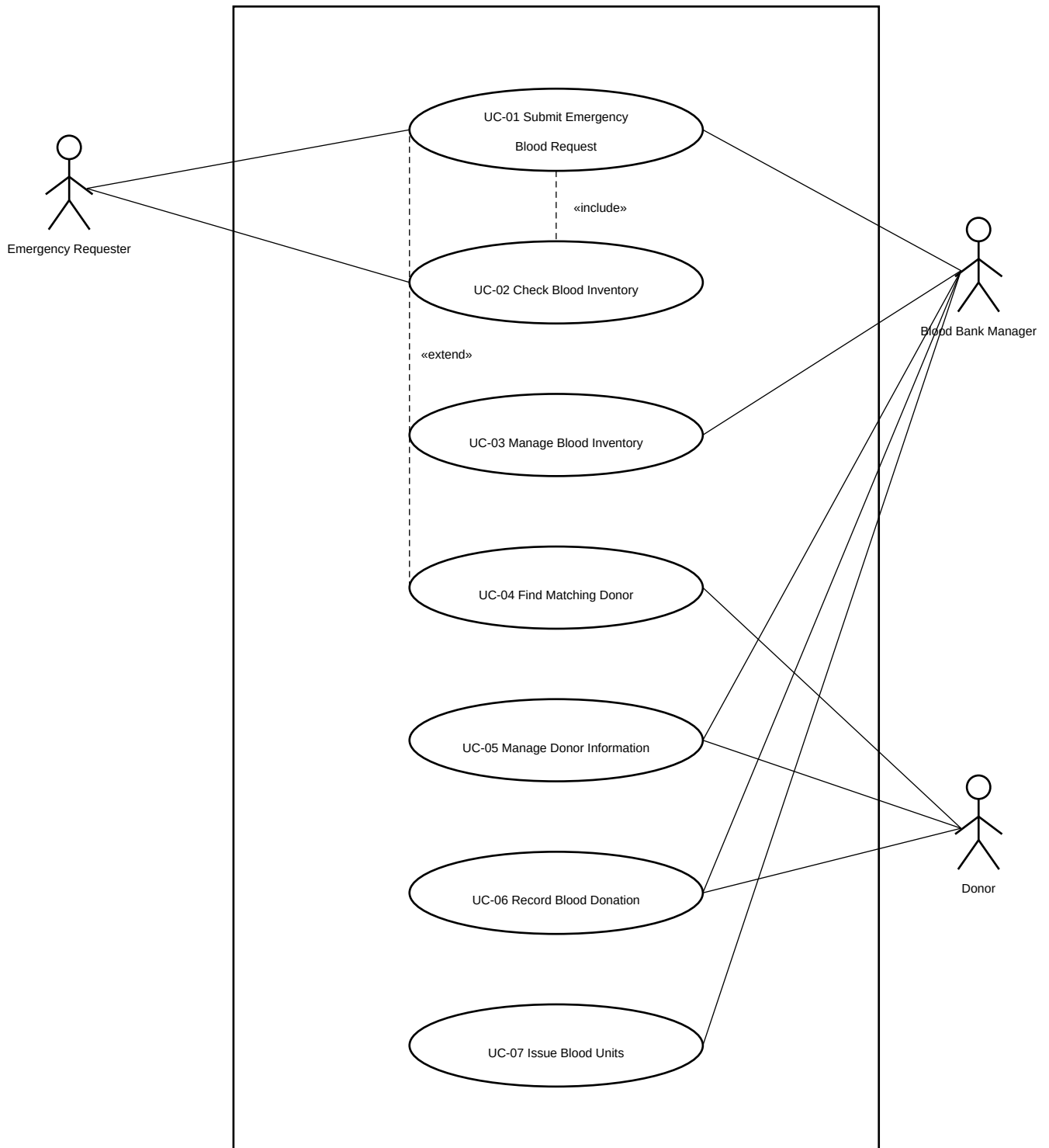
Actor	Role
Emergency Requester	Creates emergency blood requests and checks blood/request status.
Blood Bank Manager	Manages blood inventory, donor information, donations, requests and blood issue.
Donor	Registers donor details, updates availability and responds to emergency donation requests.

Primary Use Cases

ID	Use Case	Main Actor
UC-01	Submit Emergency Blood Request	Emergency Requester
UC-02	Check Blood Inventory	Emergency Requester
UC-03	Manage Blood Inventory	Blood Bank Manager
UC-04	Find Matching Donor	Blood Bank Manager / Donor
UC-05	Manage Donor Information	Blood Bank Manager / Donor
UC-06	Record Blood Donation	Blood Bank Manager / Donor
UC-07	Issue Blood Units	Blood Bank Manager

UML Use-Case Diagram

Blood Bank Inventory & Emergency Donor Matcher System



Relationships: UC-01 «include» UC-02 because inventory is checked as part of submitting a request. UC-04 «extend» UC-01 because donor matching is conditional when the required blood is unavailable or insufficient.